

8	(a)	density of gas cloud = $1.66 \times 10^{-27} \times 100 = 1.66 \times 10^{-25} \text{ kg cm}^{-3} = 1.66 \times 10^{-19} \text{ kg m}^{-3}$ [C1 if number of molecules obtained as 6.02×10^{59}] Volume of gas cloud = $\frac{10^{33}}{1.66 \times 10^{-19}} = 6.02 \times 10^{51} \text{ m}^3$ Radius of spherical gas cloud: $\frac{4}{3} \pi R^3 = 6.02 \times 10^{51}$ $R = 1.129 \times 10^{17} \text{ m} = 1.1 \times 10^{17} \text{ m}$	C1 M1 A0
	(b)	Number of molecules, $N = \frac{10^{33}}{1.66 \times 10^{-27}} = 6.02 \times 10^{59}$ Temperature T: $\frac{6.67 \times 10^{-11} \times (10^{33})^2}{1.1 \times 10^{17}} \geq \frac{3}{2} (6.02 \times 10^{59}) (1.38 \times 10^{-23}) T$ T = 43.7 K (if used 1.1×10^{17}) T = 47.4 K (if used 1.129×10^{17}) T = 48.6 K (if used more dp) T = 52.5 K (if used 1.1×10^{17} to find volume)	C1 A1
	(c)	As the density of the gas cloud increases, <u>the mass of the gas cloud increases</u> since the volume stays the same. (also accept <u>number of molecules increases</u>) From the formula, the maximum (average) <u>temperature</u> for the gas cloud to collapse will also <u>increase</u>. (Since both M and N increases, but M^2 increases more than N, thus the ratio $\frac{M^2}{N}$ increases).	M1 A1
	(d)	High kinetic energy is required to <u>overcome the (mutual electrostatic) repulsion</u> between the positively charged nuclei so that they can fuse. Or nuclei must <u>have high speed</u> to collide. Since thus <u>high kinetic energy</u> (of the nuclei) is related to temperature, thus a high temperature is required for fusion.	B1 B1

(e)	Energy released = total binding energy of products – total binding energy of reactants Energy released = $(7.07 \times 4) - (2.57 \times 3 \times 2)$ $= 12.86 \text{ MeV} = 12.9 \text{ MeV}$	C1 A1
(f)	The maximum frequency gamma photon is emitted when all the energy released in the fusion reaction is converted into the photon energy. $hf = 12.9 \text{ MeV}$ $f = \frac{12.9 \times 10^6 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$ $= 3.11 \times 10^{21} \text{ Hz}$	C1 A1
(g)	Iron has the <u>highest binding energy per nucleon</u> of any element Or Iron is the <u>most stable element</u> Hence, the fusion of iron into heavier elements <u>requires the input of energy</u> / is <u>unlikely to occur spontaneously</u>. Or There is <u>not enough energy / temperature</u> in the stars for fusion of iron.	B1 B1
(h)	gravitational force provides for centripetal force $\frac{GMm}{r^2} = \frac{mv^2}{r}$ $v = \sqrt{\frac{GM}{r}}$	C1 M1 A0
(i)	$v = \sqrt{\frac{GM}{r}}$ $v = \sqrt{\frac{G\left(\frac{4}{3}\pi r^3 \rho\right)}{r}}$ $v = \sqrt{\frac{4}{3}G\pi \rho \times r}$ <u>Since G, π, ρ are constants,</u> so $v \propto r$	C1 M1 A0

(i)	The mass is <u>uniformly distributed in a sphere nearer the galactic centre</u> and <u>increases proportionally with distance further from the galactic centre.</u> <u>OR</u> Messier 33 is likely to have a spherical distribution of mass with <u>uniform density nearer the galactic centre,</u> and <u>non-uniform density further from the galactic centre.</u> (This is because the portion of the rotation curve nearer M33's core shows a proportional relationship, similar to Fig 8.3(b) while the portion of the rotation curve further from M33's core shows an almost horizontal relationship, similar to Fig 8.3(c).)	B1 B1